

Let's talk about AI training power use

Posted by u/ExclusiveAnd • 0 points • 0 comments

Tl;dr: I did the math and it appears that training an image generation model requires approximately the same amount of energy as it does to train a human artist. This combined with the fact that AI generation is more efficient (and has enormously higher bandwidth) would suggest that AI is favorable in terms of energy *even considering training.*

I am aware that AI use is very energy efficient compared to lots of other things we could be doing (and is in fact about 100–1000x more efficient than a human performing a similar task), but also that AI training is comparatively *much* more expensive.

This isn't a huge surprise given that AI training ingests literal *billions* of training images/texts, possibly visiting each dozens or even hundreds of times. The total energy consumption ends up being measured in the [hundreds to thousands of *mega*watt-hours] (<http://arxiv.org/pdf/2211.02001>), which for large models (e.g., GPT-3 at 1300 mWh) compare to the annual consumption of around 130 homes in the US or, equivalently, 2 or 3 round-trip flights from New York to LA (so 4 or 6 total flights), or around 5 times the lifetime energy consumption of a typical sedan (assumed to be 6000 gallons of gasoline). Extrapolating based on the number of parameters, some newer models may have consumed a factor of 10 or so more than that, though it appears the current trend is for models to become smaller and more sophisticated (and thus cheaper to train). DALL·E 3 and Stable Diffusion XL, in contrast, required around 1/20 the GPU-hours to train compared to GPT-3 and so likely also 1/20 the energy.

This obviously isn't a huge amount of energy *per model*, especially considering that there are around 130M occupied homes *in the US alone*, 10M annual flights globally (though it's difficult to determine the total distance flown), and something like 100M new cars sold globally each year (which I'll use as a proxy for cars reaching the end of their life and thus "counting" their 6000 gallons). Even if 10,000 GPT-3-sized industry models are trained each year, the aggregate cost of AI training would only match around 0.1–1% of each of these industries' total power consumption each year, and I haven't even considered other massive energy-users such as manufacturing. AI is but a tiny drop in the bucket.

That said, I'm curious about AI training versus the energy cost of training a human. US citizens have amongst the highest energy consumption per capita at around 86 mWh annually. Because humans do more than just work and study, I'll only count 2000 hours per year towards "training", and I'll assume just 4 years to transition from complete beginner to

expert. This works out to about 80 mWh total energy invested in training, which, surprisingly enough, is almost identical to the estimated cost of training a top-of-the-line image generation model.

****So,**** we can gather that putting a human through art school has basically the same environmental cost as training an image generation model. Even if said humans take pains to cut their carbon footprint by 50%, we'd only get two human artists for the cost of one model, ***but** the models are then massively more efficient and, further, have essentially unbounded throughput (limited only by the number of machines running the same model).*

Power-consumption arguments in opposition to AI are not looking so hot.

Comments